

=== CONFIGURACION GNB (1780308185) ===

gNB configuration for URLLC E-Sports VR
Optimized for low latency and high reliability

Core Network (AMF/UPF) configuration

amf_addr: 10.53.1.2

amf_port: 38412

gnb_id: 0x1999 # Unique gNB ID

mcc: 001

mnc: 010

tac: 7

slices:

- sst: 1 # Slice/Service Type for URLLC
- sd: 1 # Slice Differentiator

Radio Frequency (RF) configuration

rf:
srate: 23.04e6 # Sample rate, critical for radio performance, must match UE
dl_nr_arfcn: 154000 # Downlink NR ARFCN, must match UE
ssb_nr_arfcn: 154090 # SSB NR ARFCN, must match UE
bands: [28] # NR band, must match UE
channel_bandwidth_MHz: 5 # Channel bandwidth, optimized for URLLC
tx_gain: 70 # Transmit gain
rx_gain: 40 # Receive gain
nof_antennas: 1
device_name: uhd
device_args: "type=b200,master_clock_rate=23.04e6,num_recv_frames=512,num_send_frames=512"

CU-CP configuration

cu_cp:
bind_addr: 10.53.1.3 # gNB IP, must match docker-compose ipv4_address
amf_addr: 10.53.1.2 # AMF IP
amf_port: 38412

ZMQ configuration for communication with UE (simulation)

zmq:
tx_port: 2000 # Port for transmitting to UE
rx_port: 2001 # Port for receiving from UE
bind_addr: 0.0.0.0 # Bind address for receiving

=== CONFIGURACION UE (1780308185) ===

UE configuration for URLLC E-Sports VR
Optimized for low latency and high reliability

[rf]
freq_offset = 500
tx_gain = 70
rx_gain = 40
srate = 23.04e6 # Sample rate, critical for radio performance, must match gNB
nof_antennas = 1
device_name = uhd
device_args = sync=internal

[rat.nr]
bands = 28 # NR band, must match gNB
nof_carriers = 1
max_nof_prb = 25
nof_prb = 25
dl_nr_arfcn = 154000 # Downlink NR ARFCN, must match gNB
ssb_nr_arfcn = 154090 # SSB NR ARFCN, must match gNB

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 504f20634f6320504f50206363500a4f
k = 6874736969202073796d4b2079650a73
imsi = 001010000000001 # MCC=001, MNC=010, must match gNB

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
ip_addr is explicitly forbidden here.

[gui]
enable = true

[zmq]
tx_port = 2001 # Transmit to gNB's rx_port
rx_port = 2000 # Receive from gNB's tx_port
bind_addr = 0.0.0.0 # Bind address for receiving

```
# Copyright 2023-present Intel Corporation
# SPDX-License-Identifier: Apache-2.0
```

```
version: '3.8'
```

```
services:
```

```
  5gc:
    build:
      context: .
      dockerfile: Dockerfile-5gc
    container_name: open5gs-5gc
    privileged: true
    networks:
      ran:
        ipv4_address: 10.53.1.2 # Core 5G IP
    command: 5gc -c open5gs-5gc.yml
    volumes:
      - ./configs/open5gs:/open5gs/config
      - ./logs/open5gs:/open5gs/log
    ports:
      - "3000:3000" # WebUI
      - "38412:38412/sctp" # AMF
      - "2152:2152/udp" # UPF
    cap_add:
      - NET_ADMIN
      - SYS_NICE
    healthcheck:
      test: ["CMD", "pgrep", "open5gs-amfd"]
      interval: 10s
      timeout: 5s
      retries: 5
```

```
  gnb:
    build:
      context: .
      dockerfile: Dockerfile-gnb
    container_name: srsran-gnb
    privileged: true
    networks:
      ran:
        ipv4_address: 10.53.1.3 # gNB IP, must match gNB bind_addr
    command: gnb -c srsran-gnb.yml
    volumes:
      - ./configs/srsran:/srsran/config
      - ./logs/srsran:/srsran/log
    cap_add:
      - NET_ADMIN
      - SYS_NICE
    healthcheck:
      test: ["CMD", "pgrep", "srsran_gnb"]
      interval: 10s
      timeout: 5s
      retries: 5
```

```
networks:
```

```
  ran:
    driver: bridge
    ipam:
      config:
        - subnet: 10.53.1.0/24
```